

Model Fitting & Model Comparison

Reflective summary of the model-fitting stage | Clinical treatment outcome classification

Executive Reflection

The model-fitting stage translated the cleaned and feature-engineered clinical data into comparable predictive evidence. I used the same leakage-safe stratified 60/20/20 train/validation/test split for every model, tuning and selecting hyperparameters from training/validation data only and touching the test set exactly once per model. This made the comparison fair and kept the test set representative of the real class distribution.

Models compared	Selection priority	Best current model
7	PR-AUC, then F1-Score	XGBoost

Modelling Decisions

Class imbalance (~22% Effective) was handled via `scale_pos_weight` (tree-based models) or `class_weight='balanced'` rather than resampling (no SMOTE), so no synthetic rows were introduced. Because the engineered dataset contains categorical variables, a train-only-fitted frequency-rank encoder converted them to numeric codes (median imputation for numeric columns); the untouched validation/test partitions were transformed with those same fitted statistics. This sequence reduces the risk of target leakage.

Seven models were intentionally compared using the same held-out test data: Decision Tree, Random Forest, XGBoost, Gradient Boosting, LightGBM, LSTM, and an MLP. The tree-based models were configured with constrained depth and minimum sample requirements where specified, balancing predictive flexibility against overfitting risk.

Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
XGBoost	0.5817	0.3025	0.6683	0.4165	0.6545	0.3465
Gradient Boosting	0.5868	0.3043	0.6609	0.4168	0.6545	0.3464
MLP	0.5537	0.2946	0.7158	0.4174	0.6540	0.3454
LSTM	0.5712	0.2990	0.6842	0.4162	0.6540	0.3452
Random Forest	0.5702	0.2986	0.6852	0.4160	0.6526	0.3429
Decision Tree	0.5385	0.2878	0.7234	0.4118	0.6473	0.3320
LightGBM	0.5451	0.2865	0.6955	0.4059	0.6345	0.3164

Interpretation

The current comparison ranks **XGBoost** first with PR-AUC 0.3465 and F1-Score 0.4165. PR-AUC (average precision) was used as the primary ranking metric because, unlike ROC-AUC or accuracy, it is far more sensitive to performance on the minority (Effective) class in an imbalanced dataset; F1-Score provides a useful secondary check on the precision/recall balance.

This result is a model-selection signal rather than evidence that the model is clinically ready. A strong score on one fixed split can still be affected by sampling variation, feature leakage, preprocessing assumptions, or changes in the population. The model should therefore be stress-tested with repeated cross-validation, calibration analysis, threshold analysis, and an external or temporal validation set before deployment.

Reflection on Limitations

Frequency-rank encoding is convenient for these tree/boosting models, but its numerical ordering is not inherently meaningful and neural nets (LSTM/MLP) additionally need feature scaling to train stably. Imputation can also make records appear more complete than the underlying data really are. These choices were appropriate for a consistent baseline comparison, but they should be documented and revisited during model review.

Next Steps

The next stage should confirm that the selected model remains stable across folds and clinically meaningful subgroups. I would also inspect calibration, precision-recall trade-offs, false-negative cases, feature importance stability, and the effect of alternate decision thresholds. Only after those checks should a final model and operating threshold be considered for a controlled evaluation setting.

Source: per-model metrics JSON files in `metrics_output/`. Model artifacts (.pkl/.h5) are stored alongside them.